

# Twishha Soni

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## SUMMARY

Recent graduate in **Computer Engineering**, building at the intersection of **backend engineering** and **AI** — with hands-on experience in **REST API design** (Java, Spring Boot) and end-to-end **AI systems** (RAG pipelines, LLM integration, MCP). During a **7-month internship**, helped a school staff team **adopt AI tools** into their daily workflows — translating real operational problems into practical **no-code and low-code solutions**, and communicating technical ideas clearly to non-technical people. Strong at figuring out where AI actually fits a use case and getting others comfortable using it. Open to **internship opportunities in backend or AI engineering**.

## EXPERIENCE

### Intern — AI & Tools Application

*Podar International School*

Sep 2025 – Mar 2026

*Modasa, Gujarat (On-site)*

- Joined during a **subject teacher crisis** mid-semester — built a **Notion workspace** with **Notion AI** letting each teacher track **syllabus progress** independently, giving coordinators and the principal **real-time visibility** to measure and recover learning loss.
- Introduced **Claude AI** to teaching staff and ran **hands-on sessions** for creating daily worksheets and classroom materials, cutting prep time and freeing teachers to focus on students.
- Built a **Google Apps Script** attendance tracker inside **Google Sheets** using Claude for Excel for the school's **English Leveling-Up Program**, giving staff a clean, no-fuss tool to manage attendance across different student level groups.

## PROJECTS

### 1. ReelMind — Semantic Instagram Personally Saved Reel Search

[GitHub Repo](#)

*Python · python-telegram-bot · yt-dlp · Gemini (multimodal) · ChromaDB · sentence-transformers · FastAPI · Docker*

- Solving a real gap in Instagram's UX: **saved reels have no native search**, and third-party applications are web based — **built a Telegram bot instead**, turning your phone's chat interface into a zero-friction reel search assistant available wherever Instagram is.
- Indexes reels from Instagram URLs or bulk JSON export: **downloads video via yt-dlp**, uploads to **Gemini File API overfor multimodal analysis**, generates **rich semantic summaries**, embeds them with sentence-transformers, and stores in ChromaDB — **full RAG pipeline** over video content.
- Distributes API load across two Gemini models (video understanding vs. answer formatting) to stay within free-tier quota limits; config-centralized model names for one-line swaps. **Planned: swap video model for locally hosted Gemma 4 via Ollama to eliminate free-tier latency and API rate limits.**
- Containerized and deployed** it on **AWS EC2 via Docker and Docker Compose**, eliminating environment-parity issues from a prior deployment by isolating dependencies and **persisting ChromaDB on a separate EBS volume**.

### 2. Doc-IQ — Document Intelligence Assistant

[GitHub Repo](#)

*Python · FastAPI · ChromaDB · sentence-transformers · Google Gemini · FastMCP · AWS EC2 · Ubuntu · systemd*

- Built a full-stack **RAG application** where users upload **.pdf/.txt** documents and ask **natural-language questions** — the system retrieves the most semantically relevant chunks and returns **cited, grounded answers** via **Google Gemini**, eliminating manual document searching.
- Deployed on AWS EC2** (c7i-flex.large, Ubuntu) with **systemd managing automatic startup on reboot**; **mounted an EBS gp3 volume at /data for persistent ChromaDB storage that survives instance restarts.**
- Exposed two interfaces — a **FastAPI** web server with a zero-dependency HTML/CSS/JS frontend, and a **FastMCP server** for direct **Claude Desktop** integration; built across clean, dedicated ingest, embed, retrieve, and generate layers.

## TECHNICAL SKILLS

**Languages:** Python, Java (17), SQL

**AI & ML:** RAG pipelines, Prompt Engineering, LLM integration (Gemini, Claude), sentence-transformers, ChromaDB, FastMCP (Model Context Protocol)

**Frameworks & Libraries:** FastAPI, Spring Boot, Spring Security, Spring Data JPA, Hibernate, JWT

**Databases:** MySQL, ChromaDB, H2

**Tools & Platforms:** Git, Postman, Docker, Linux, Maven, IntelliJ IDEA, Google Apps Script, Notion

**Concepts:** REST API Design, JWT Auth, Role-Based Access Control, OOP, SOLID Principles, ManyToMany JPA, Global Exception Handling, Retrieval-Augmented Generation(RAG), Model Context Protocol(MCP)

## EDUCATION

**Gujarat Technological University (GTU)**

*Bachelor of Engineering in Computer Engineering | CGPA: 7.9*

Modasa, Gujarat

*Sept 2022 – May 2026*